

Run 11 | OME

India

IV + controls

Target Opportunity-motivated entrepreneurship; binary 1 vs. 0
Sample 27,555 respondents **Coverage** 1 country; 2015–2022
Positive cases 2,167 (7.86%) **Case time** 1m 01s
Train / val. / test 18,739 / 3,306 / 5,510 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 26.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: H2O GBM (0.9877). Highest F1: Ridge (0.7943). For H2O GBM, adding controls changed ROC-AUC by +0.0005 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
H2O GBM	0.9877	0.8565	96.19	70.46	88.68	0.7853	4s
Random Forest	0.9876	0.8582	96.41	75.05	81.29	0.7805	1s
CatBoost	0.9875	0.8523	96.08	73.33	78.75	0.7595	7s
XGBoost	0.9872	0.8464	96.33	71.51	88.68	0.7918	<1s
Ridge	0.9868	0.8461	96.32	70.76	90.53	0.7943	11s
Lasso	0.9868	0.8445	96.17	70.33	88.68	0.7845	10s
Logistic Regression	0.9867	0.8441	96.23	70.34	89.84	0.7890	2s
Elastic Net	0.9863	0.8390	96.17	70.33	88.68	0.7845	15s
Decision Tree	0.9845	0.7893	95.63	67.78	84.53	0.7523	<1s
Naive Bayes	0.9095	0.5607	93.21	56.26	61.20	0.5863	7s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	750.28	40.96
2	Established business owner	256.76	14.02
3	Business angel	74.85	4.09
4	Occupational status	72.09	3.94
5	Knowing entrepreneurs	61.03	3.33
6	Age	60.14	3.28
7	Household size	45.07	2.46
8	Business exit	40.49	2.21

Share is the percentage of total forest impurity importance across all 66 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Living conditions	11.92	0.65
2	Investment environment	10.25	0.56
3	Enterprise conditions	9.96	0.54

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 2 batch, 28 September 2026; 11_obj2_india_iv_cv_OME. Scores and importance are taken directly from saved CSV outputs.